

SokaSoko — REFEREE Subscription Test Script (Internal)

Version 1.0 · Internal QA — includes admin path and backend enforcement

What you're testing

REFEREE subscription with an activity-based free trial. Refs officiate free until 10 COMPLETED matches (main ref OR either assistant slot). At game 8 a 'karibu' notification fires; at game 10 they are blocked from new assignments unless subscribed. Existing refs already at threshold receive a one-time 15-day grandfather window. Server picks MINOR vs ADULT tier from DOB. This doc includes admin activation and backend enforcement points.

Prerequisites

- Android device on the latest internal-testing build.
- A REFEREE test account (ideally one with < 5 completed matches).
- An admin account for tier activation and Notification inspection.
- A team-owner account (ACADEMY/CLUB/SCHOOL/COACH) that can schedule matches with this ref.
- Ability to seed match rows or run completions to hit game milestones (or MongoDB access).

Environment

- Backend: <https://sokasoko.onrender.com> (verify `/v1/subscriptions/catalog` returns REFEREE MINOR/ADULT under prices+features).
- Backend commit: 68f189b or later on main.
- Mobile: sokasoko-mobile main branch commit 7130623 or later.
- Constants baked in: REFEREE_FREE_GAME_THRESHOLD=10, REFEREE_WARN_AT_GAMES=8, REFEREE_GRANDFATHER_DAYS=15.

Test cases

1. Menu visibility for REFEREE

- Log in as REFEREE, open Profile, tap ■.

Expected: 'Uandikishaji' entry appears at the top of the menu (previously PLAYER-only).

2. Fresh REFEREE — free trial state

- Open Uandikishaji on a REFEREE with 0 completed matches.
- Confirm GET `/v1/subscriptions/me?userId=&userType;=REFEREE` payload.

Expected: `refereeStatus = { eligible:true, reason:'FREE_TRIAL', gamesOfficiated:0, subscribed:false, tier:null, grandfatherEndsAt:null, threshold:10, warnAt:8, ageBracket:'ADULT'|'MINOR' }`. Screen shows 'Bado uko bure', progress bar 0/10, preferential-selection banner, and plan cards for the correct age bracket.

3. Age-bracket pricing

- Test with a REFEREE whose DOB makes them under 18.
- Test with a REFEREE whose DOB is 18+.
- Test with a REFEREE whose DOB is null.

Expected: MINOR (<18): Monthly TSh 5,000; Quarterly TSh 10,000; no Biannual. ADULT (>=18) and null DOB: Monthly 10,000 / Quarterly 25,000 / Biannual 40,000. Client does not present a bracket picker — bracket is server-authoritative.

4. Anytime subscribe — pre-threshold

- As a REFEREE with 0-9 completed matches, tap a plan card.
- Submit through the payment sheet with reference TESTREF001.

Expected: POST /v1/subscriptions returns 201 with tier=MINOR|ADULT (auto-derived), status=PENDING. Screen snackbar: 'Malipo yamepokelewa...'. Screen reloads; state still FREE_TRIAL until admin activates.

5. Admin activation

- As admin, GET /v1/subscriptions?status=PENDING&userType;=REFEREE.
- POST /v1/subscriptions/:id/activate.

Expected: Subscription flips to ACTIVE with startDate, endDate, gracePeriodEndsAt computed from the plan. Referee's /me now returns refereeStatus.reason='SUBSCRIBED'.

6. Warning notification at game 8

- Set the REFEREE's completed-match count to 7 (via DB seed, or run 7 real completions).
- Complete an 8th match with this REFEREE in referee OR assistantReferee1/2.
- Check the ref's Notification collection.

Expected: One new Notification created with type='SUBSCRIPTION', title='Karibu utahitajika kujisajili', body includes 'mechi 8' and 'mechi 2'. Only fires if the ref is NOT currently subscribed. Subscribed refs receive NO notification.

7. Threshold notification + block at game 10

- Complete the 10th match with an unsubscribed ref.
- Attempt POST /v1/matches with this ref as referee/assistant on a new match.

Expected: Notification: 'Uandikishaji unahitajika' with body about 15-day grace. The new /matches POST returns 403 with reason='REFEREE_SUBSCRIPTION_REQUIRED' and error mentions '(mechi 10)'. NOTE: current code stamps grandfatherUntil on first eligibility check post-threshold, so the FIRST attempted assignment might still succeed because grandfather window applies. Second attempt after 15+ days will be blocked.

8. Grandfather window

- Trigger getRefereeEligibility for a ref at >=10 games with no subscription (first time).
- Inspect User.refereeGrandfatherUntil in DB.

Expected: First call stamps refereeGrandfatherUntil = now + 15 days on the User doc. During this window, eligible=true, reason='GRANDFATHER'. After it passes, eligible=false, reason='SUBSCRIPTION_REQUIRED'.

9. Post-completion hook fires for all three slots

- Complete a match where the ref is in assistantReferee2 (not head ref).

Expected: The 8/10 notifications fire for assistantReferee2 as well — the hook considers all three slots. Count aggregation Match.countDocuments should match this.

10. GET /v1/users?type=REFEREE enrichment

- GET /v1/users?type=REFEREE&limit;=20
- GET /v1/users?type=REFEREE&limit;=20&sortBySubscribed;=1

Expected: First call: each row has refereeSubscribed boolean. Second call: rows are sorted with refereeSubscribed=true first, then =false. Order within each group falls back to createdAt desc from the original query.

11. Server auto-picks MINOR/ADULT

- POST /v1/subscriptions with userType=REFEREE and tier=STANDARD in body.

Expected: Backend ignores the client tier and re-derives from user.dob. Response tier is MINOR or ADULT. This prevents a client from smuggling a cheaper tier for an adult ref.

12. Non-referee cannot use REFEREE tier catalog

- POST /v1/subscriptions with a PLAYER user, userType=REFEREE.

Expected: Since userType is derived from body not user record, this is not fully prevented server-side. Regression check only — a UI/CSRF guard is a follow-up. Log any misuse.

13. Existing enforcements not broken

- PLAYER subscription still works (case 1-15 in the PLAYER doc).
- PLAYER Ismaili quota still enforces.

Expected: PLAYER flow unchanged. No regression in Ismaili / scout-reports / eval-requests.

14. Bell notification pipeline

- Open the Notifications screen after case 6 or 7.

Expected: Referee sees the newly-created SUBSCRIPTION notifications in their inbox.

15. Overflow / typography audit

- Test on narrow devices (Pixel 4a) — REFEREE main screen.

Expected: Progress bar row fits. Plan cards do not overlap. Payment sheet keyboard push works. Grandfather countdown fits without ellipsis.

How to report a bug

Field	Example
Test case #	7 — Threshold block at game 10
Device / OS	Pixel 6a · Android 14
App version	1.0.1+14
Backend commit	68f189b
Account (type + number)	REFEREE · TFH-R-A000012
gamesOfficiated (from /me)	10
Endpoint (if backend)	POST /v1/matches
Expected	403 REFEREE_SUBSCRIPTION_REQUIRED
Actual	201 — match created without gate
Screenshot / repro	attached
Render log excerpt	attached

Known limitations (not bugs)

- Payment gateway is not wired; paid subscriptions stay PENDING until admin activation.
- Placeholder phone numbers in `_paymentInstructions()` must be replaced with real merchant numbers before external testers use this.
- No mid-grace SMS reminder cadence yet — only bell notifications on transition into GRACE and EXPIRED (subscription lapse) and on hitting games 8 & 10 (referee).
- Referee-picker in the schedule-match screen does not yet visually badge subscribed refs; backend enrichment is ready.
- Regression: if a REFEREE grew past 10 games during pre-deploy testing, the very first eligibility check after deploy grants the grandfather window; test freshly seeded users to observe the block path.
- Server-side check does not currently prevent using `userType=REFEREE` for a PLAYER user body — add UI/session guard as follow-up.